

## Worksheet 16 - Calculating silhouette width

Study the code below (requires Cluster package):

```
> SepalLength<-c(5.1, 4.9, 7.0, 5.0, 5.1, 6.3, 6.5, 6.9)
> PetalLength<-c(1.4, 1.5, 4.7, 3.5, 3.3, 6.0, 5.1, 5.7)
> df<-data.frame(SepalLength, PetalLength)
> result<-pam(df, 3, metric="euclidean")
> result
```

Medoids:

|      | ID | SepalLength | PetalLength |
|------|----|-------------|-------------|
| [1,] | 2  | 4.9         | 1.5         |
| [2,] | 7  | 6.5         | 5.1         |
| [3,] | 5  | 5.1         | 3.3         |

Clustering vector:

```
[1] 1 1 2 3 3 2 2 2
```

Objective function:

|  | build     | swap      |
|--|-----------|-----------|
|  | 0.3413238 | 0.3413238 |

```
> silhouette(result)
```

|   | cluster | neighbor | sil_width |
|---|---------|----------|-----------|
| 1 | 1       | 3        | 0.8882631 |
| 2 | 1       | 3        | 0.8827312 |
| 8 | 2       | 3        | 0.7294776 |
| 7 | 2       | 3        | 0.6597197 |
| 6 | 2       | 3        | 0.6455286 |
| 3 | 2       | 3        | 0.5564830 |
| 4 | 3       | 1        | 0.8910531 |
| 5 | 3       | 1        | 0.8794922 |

For your convenience, the clustering result is rewritten in the following format (please verify):

| <u>Cluster 1</u> |              | <u>Cluster 3</u> |              | <u>Cluster 2</u> |              |
|------------------|--------------|------------------|--------------|------------------|--------------|
| Sepal Length     | Petal Length | Sepal Length     | Petal Length | Sepal Length     | Petal Length |
| 5.1              | 1.4          | 5.0              | 3.5          | 6.3              | 6.0          |
| 4.9              | 1.5          | 5.1              | 3.3          | 6.5              | 5.1          |
|                  |              |                  |              | 6.9              | 5.7          |
|                  |              |                  |              | 7.0              | 4.7          |

To confirm you understand how to calculate silhouette widths, validate through hand calculations that the silhouette width for the observation with SepalLength 4.9 and PetalLength 1.5 in cluster 1 is 0.8827312 . Use Euclidean distance for your calculations.